PLSC 48110/28110

Lecture 6: Power calculations and randomization procedures

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One-pager overview

- See the syllabus for the submission deadline.
- Meet with me to discuss; schedule a meeting at https://calendly.com/mollyow/office-hours
- One-pager should cover general research question, how you would think about answering it, and some general design considerations. You are not directly graded on the one-pager, but it allows us to give you feedback to make your final project better.
- See Lecture 5 for design documents and pre-registrations.

Considerations when running an experiment

- What hypotheses are you testing? And what estimation techniques used to test them?
- What proportion of subjects in treatment vs. control? Or, with multiple conditions, what proportion in each?
- Oversampling some sub-populations?
- (Replicable) methods for randomization (is your study online? in person? do you have the full sample frame ex ante?)
- Also: ethical considerations, cost, implementation barriers

Sample frame

- Who is included in your study? How are they selected?
- Random selection? Clustered selection ? (e.g., schools, neighborhoods) Somehow representative of a population? Convenience sample? Eligibility criteria?
- Important for both internal and external validity.
- Approaches to creating a sample frame:
  - Administrative data (census, school registers, health data)
  - Do your own census/mapping exercise
  - "Random walk"
  - Ongoing enrollment: online population, people at a health clinic; generally have eligibility criteria

Randomization procedures

- Simple random assignment
- Complete random assignment
- Stratified random assignment
- Paired random assignment
- Clustered random assignment

Randomization procedures: mechanisms

- Is it replicable? Is it public? What could go wrong?
- Lottery
- Random walk
- Computer program

Simple random assignment

- Treatment assignment probabilities are the same across units
- Treatment is assigned to each unit independently
- Not optimal for estimation $\rightarrow$ can have chance imbalance in treatment and control.
- But sometimes required in practice, e.g., when you don't have your entire sample frame prior to starting the experiment

Complete random assignment

- Fix the number/proportion of units in treatment and control, then randomize over sample frame.
- We can still get chance imbalance on measured and unmeasured variables.

Stratified random assignment

- Divide your sample frame into subgroups. Apply complete random assignment *within* subgroups.
- Benefits: removes chance of imbalance on *measured* variables. Especially useful if we want to look at treatment effects within subgroups.
- Sometimes different strata have different assignment probabilities.

Stratified random assignment

- How do you decide what variables to stratify on, and where to bin continuous variables?
- Variable should correlate with outcome variable.
- For binning, can use statistical methods (e.g., decision trees).

Clustered random assignment

- Divide your sample into subgroups.
- Assign treatment uniformly within subgroups. (Or at fixed levels of treatment intensity.)
- Bias-variance trade-off.
- Useful when there is spillover, and/or you're interested in what a program would look like if implemented uniformly at scale.
- Setting different levels of treatment intensity can facilitate estimating spillovers as well as direct effects. (Baird et al., 2018)

What about re-randomization?

- What are criteria?

Power

- How large does my experiment need to be for my proposed statistical procedure to be informative? (under certain conditions)
- The test can be about distinguishing treatment effects from zero, but it doesn't *have* to be.
- Hypothesis: alternative vs. null.

Power

- Type I error: falsely reject the null. Probability in terms of α.
- Type II error: fail to detect an effect when it exists. Probability in terms of β.
- Power: probability of rejecting a "false null hypothesis." I.e., given that there *is* an effect, do you correctly reject the null? Probability in terms of $1 - \beta$. Often, we target power of $1 - \beta$.
- Minimum detectable effect: the smallest effect that would produce our target power, given a specific statistical test and a given α. Selection often informed by substantive knowledge or policy relevance.

Power

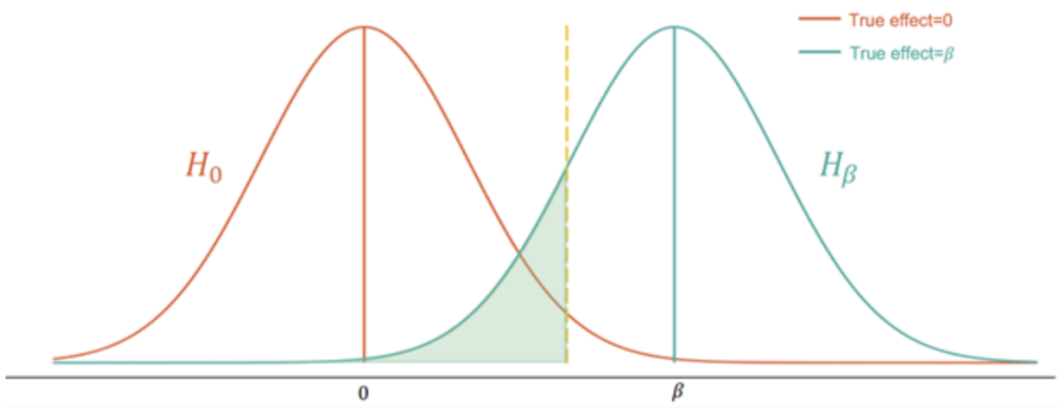


Figure 1

Power

Note: power is with respect to:

- Statistical elements: a given statistical test, a given α, a given minimum detectable effect size, a given sample size. We also need to specify the statistical distribution of our response variable.
  - Distributions: normal, binomial. Intra-cluster correlation?
- Design elements: sample size, treatment allocation.

Example

```
> library(ggplot2)
> library(estimatr) # for lin estimator
> set.seed(60637) # setting a seed is important
> complete_assign ← function(n){
+ nt ← n %/% 2 + rbinom(1, 1, n %% 2)
+ sample(rep(c(0, 1), c(n - nt, nt)))
+ }
```

Example

You can use the `randomizr` package in R. It has functions like `complete_ra()` to implement complete random assignment.

Example

Suppose:

- We have $n = 100$ observations in our sample.
- We assign treatment W to half of the sample using complete random assignment.
- There is an X variable distributed $\mathcal{N}(0, 1)$
- The treatment effect τ is equal to 2.
- Individual response follows:

$$Y = 3X + \tau \times W + \epsilon$$

Where ϵ is random noise distributed $\mathcal{N}(0, 1)$.

Example

```
> n ← 1e2
> W ← complete_assign(n)
> X ← rnorm(mean = 0, sd = 1, n = n)
> tau ← 2
> Y ← 3*X + tau*W + rnorm(mean = 0, sd = 1, n = n)
```

Example

We consider both X, and a dichotomized variable $\tilde{X}$, which is 1 when $X > 0$ and 0 if $X < 0$.

```
> Xtilde ← 1*(X>0)
```

```
> dat ← data.frame(Y, W, X, Xtilde)
> head(dat)
```

```
           Y W           X Xtilde
1 -0.3918746 1 -0.29282895      0
2  0.1056477 1 -0.49140540      0
3  1.5193091 0  0.15595692      1
4  0.5207658 0  0.08337197      1
5  5.8685348 0  2.31585521      1
6  2.0938111 1  0.04878741      1
```

Example

We can estimate our treatment effect as a difference in means.

```
> difference_in_means(Y ~ W, data = dat) # from estimatr
```

```
Design:  Standard
  Estimate Std. Error  t value  Pr(>|t|)   CI Lower CI Upper       DF
W 1.264531  0.6129927 2.062881 0.0417812 0.04799564 2.481067 97.54236
```

```
> mean(Y[W == 1] - mean(Y[W == 0])) # equivalent
```

```
[1] 1.264531
```

Example

- How is precision affected if we had used a different design? A different estimator?
- We will consider simple random assignment, complete random assignment, stratified random assignment, and paired random assignment.

Example

```
> niter ← 1e4
> estimates ← estimates_alt ← power ← power_alt ← matrix(NA,
+ nrow = niter,
+ ncol = 4)
> colnames(estimates) ← colnames(estimates_alt) ← colnames(
    power) ←
+ colnames(power_alt) ← c('simple', 'complete', 'stratified',
     'paired')
```

Simulation: Simple random assignment

```
> set.seed(60637)
> for (i in 1:niter) {
+ X ← rnorm(mean = 0, sd = 1, n = n)
+ Xtilde ← 1*(X>0)
+
+ # Simple random assignment
+ W ← sample(0:1, size = n, replace = TRUE)
+ Y ← 3*X + tau*W + rnorm(mean = 0, sd = 1, n = n)
+
+ est0 ← difference_in_means(Y~ W, data = data.frame(Y, W))
+ est1 ← lm_lin(Y~ W, covariates = formula('~ Xtilde'),
+ data = data.frame(Y, W, Xtilde))
+ estimates[i, 1] ← coef(est0)
+ power[i, 1] ← (est0$conf.low > 0 | est0$conf.high < 0)
+ estimates_alt[i, 1] ← coef(est1)[['W']]
+ power_alt[i, 1] ← (est1$conf.low['W'] > 0 | est1$conf.high[
    'W'] < 0)
+ }
```

Simulation: Complete random assignment

```
> set.seed(60638)
> for (i in 1:niter) {
+ X ← rnorm(mean = 0, sd = 1, n = n)
+ Xtilde ← 1*(X>0)
+
+ # Complete random assignment
+ W ← complete_assign(n)
+ Y ← 3*X + tau*W + rnorm(mean = 0, sd = 1, n = n)
+
+ est0 ← difference_in_means(Y~ W, data = data.frame(Y, W))
+ est1 ← lm_lin(Y~ W, covariates = formula('~ Xtilde'),
+ data = data.frame(Y, W, Xtilde))
+ estimates[i, 2] ← coef(est0)
+ power[i, 2] ← (est0$conf.low > 0 | est0$conf.high < 0)
+ estimates_alt[i, 2] ← coef(est1)[['W']]
+ power_alt[i, 2] ← (est1$conf.low['W'] > 0 | est1$conf.high[
    'W'] < 0)
+ }
```

Simulation: Stratified random assignment

```
> set.seed(60639)
> for (i in 1:niter) {
+ X ← rnorm(mean = 0, sd = 1, n = n)
+ Xtilde ← 1*(X>0)
+
+ # Stratified random assignment
+ W ← rep(NA, n)
+ W[which(Xtilde == 0)] ← complete_assign(sum((Xtilde == 0)))
+ W[which(Xtilde == 1)] ← complete_assign(sum((Xtilde == 1)))
+ Y ← 3*X + tau*W + rnorm(mean = 0, sd = 1, n = n)
+
+ dat ← data.frame(Y, W, Xtilde)
+ est0 ← difference_in_means(Y~ W, blocks = Xtilde, data =
    dat)
+ est1 ← lm_lin(Y~ W, covariates = ~ Xtilde, data = dat)
+ estimates[i, 3] ← coef(est0)
+ power[i, 3] ← (est0$conf.low > 0 | est0$conf.high < 0)
+ estimates_alt[i, 3] ← coef(est1)[['W']]
+ power_alt[i, 3] ← (est1$conf.low['W'] > 0 | est1$conf.high[
    'W'] < 0)
+ }
```

Simulation: Paired random assignment

```
> set.seed(60640)
> for (i in 1:niter) {
+ X ← rnorm(mean = 0, sd = 1, n = n)
+ Xtilde ← 1*(X>0)
+
+ # Pair adjacent units in sorted X
+ pair ← integer(n)
+ pair[order(X)] ← rep(seq_len(n/2), each = 2)
+ W[order(X)] ← as.vector(replicate(n/2, sample(0:1)))
+ Y ← 3*X + tau*W + rnorm(mean = 0, sd = 1, n = n)
+
+ dat ← data.frame(Y, W, Xtilde, pair)
+ est0 ← difference_in_means(Y~ W, blocks = pair, data = dat)
+ est1 ← lm_lin(Y~ W, covariates = ~ Xtilde,
+ clusters = pair, data = dat)
+ estimates[i, 4] ← coef(est0)
+ power[i, 4] ← (est0$conf.low > 0 | est0$conf.high < 0)
+ estimates_alt[i, 4] ← coef(est1)[['W']]
+ power_alt[i, 4] ← (est1$conf.low['W'] > 0 | est1$conf.high[
    'W'] < 0)
+ }
```

Matching inference to the design

- For stratified and paired assignment, pass the stratum or pair identifier to `difference_in_means()` using `blocks`.
- The Lin regression includes centered $\tilde{X}$ and its interaction with treatment. For stratification on $\tilde{X}$, this fits a separate treatment effect within each stratum.
- For paired assignment, use pair-clustered standard errors in the Lin regression to account for dependence within pairs.
- Power below is the rejection rate of a two-sided test at $\alpha = 0.05$ when the true effect is 2.

Simulation results: variation

Variation of the estimate

```
> apply(estimates, 2, sd) # using difference in means
    estimate
```

```
   simple   complete stratified     paired
0.6410817  0.6318874  0.4107896  0.2069686
```

```
> apply(estimates_alt, 2, sd) # using lin estimator
```

```
   simple   complete stratified     paired
0.4183358  0.4147702  0.4107896  0.2165782
```

Simulation results: power

```
> apply(power, 2, mean) # using difference in means estimate
```

```
   simple   complete stratified     paired
   0.8768     0.8772     0.9976     1.0000
```

```
> apply(power_alt, 2, mean) # using lin estimator
```

```
   simple   complete stratified     paired
   0.9981     0.9963     0.9976     1.0000
```

How do you conduct power in practice?

- Determine the appropriate minimum detectable effect size.
- Determine your estimating procedure.
- Posit a response distribution.
- Simulate over different sample sizes.
- OR, if simple, use an online power calculations calculator.

References I

Baird, S., Bohren, J. A., McIntosh, C., and Özler, B. (2018). Optimal design of experiments in the presence of interference. Review of Economics and Statistics, 100(5):844–860.